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Text summarization techniques using LLMs on patients interview transcripts post pediatric heart surgery

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1 Introduction

The high risk project I chose is to summarize long texts using LLMs. In today's times where there is an abundance of text generation, I believe summarization of text is an important task. Summarization helps save time, improves readability and ignores unwanted information. It helps extract the main ideas of a text in a meaningful way. As humans, we have the cognitive ability to understand text and produce a summary of main points. Whereas, computers don't have the ability as humans which makes text summarization a very challenging problem. Text summarization can be useful in many areas. For example, in healthcare thousands of patient records are submitted to the insurance providers everyday and it is challenging for providers to process these records as quickly as possible. In such instances, extracting only the key points from the extensive medical record and generating a brief version of the record written in an easy-to-understand language can help reviewers and auditors to process these records quickly. Other examples include news websites that produce short descriptions of news topics usually as headlines and quick recaps. For this project, I used interview transcripts of patients and their parents post pediatric heart surgery to summarize topics discussed, emotions of patients and medical gaps.

2 Related Work

Text summarization in the early days was mostly rules based. Some techniques used were ranking different parts of a text according to their importance, key phrase extraction from text and sentence scoring. Baeza-Yates and Ribeiro-Neto's work on information retrieval [1] made some significant progress. The proposed methodology was tf-idf (Term Frequency - Inverse Document Frequency), a vocabulary of "words" is generated and for each document in the corpus, a count is populated based on the word frequency in the document to inverse document frequency. The end result is a term-document matrix with tf-idf values for each word of the documents in a corpus. Some drawbacks of this approach are that it does not capture semantic meaning of the words. This limits the generation of meaningful text summaries. And the tf component of the matrix can be affected by documents of different lengths which could lead to bias too.

To address these drawbacks, there have been several proposals of dimensionality reduction techniques like Gong and Liu proposed a method using latent semantic analysis (LSA) to select highly ranked sentences for document summarization in the news domain. In this method, a word-sentence matrix is built, where each row corresponds to a word from the input and each column corresponds to a sentence [2]. Each entry of the matrix is the weight of the word in the sentence. The weights of the words are computed using tf-idf and if a sentence does not have a word, the weight of that word is zero. After that, singular value decomposition (SVD) is used to transform the matrix to indicate how much a sentence represents a topic. This is indicated by weights of the topic in a sentence. Gong and Liu's method was to choose one sentence per topic and put them all together to get the final summary. And based on how long the summary is in terms of sentences, they retained the number of topics. This strategy does not account for topics that may need more than one sentence to represent its information.

Eventually, probabilistic topic models gained at-

tention. Latent Dirichlet Allocation (LDA) model is a powerful unsupervised technique for extracting information from a corpus of documents. The main idea is that it groups words and documents into groups with the notion that documents in a topic will have a lot of words in common. Also, LDA is a bag-of-words model indicating that it only considers the words and not their relationship in a sentance. David M. Blei, Andrew Y. Ng and Michael I. Jordan's paper on Latent Dirichlet Allocation LDA [3], a generative probablistic model uses probabilities to represent a document. Celikyilmaz et al. [4] describe multi-document summarization as a prediction problem based on a two-phase model. First, they propose a hierarchical topic model that identifies topic structures of all sentences. They use human generated summaries and a tree-based sentence scoring function to compute summaries. Then, use the scores and train a regression model based on the structural characteristics of the sentences and use this model to score sentences of unseen documents. Finally use the highest scoring sentences to form a summary.

Most recently with the availability of LLMs like ChatGPT, it has become very easy to summarize long texts. There are a lot of conventional methods of using LLMs to summarize long texts like recursive summarization using map-reduce etc. Sumit, Sagil, Pengcheng and Aaron's paper [5] on summarizing meeting notes based on user context using LLMs talk about improving from natural interactions to improve personal relevance and summarization quality.

3 Methodology

I first tried the basic approach of splitting the text into chunks, summarizing each chunk using LLM and produce a final summary by generating summary of the concatenated summaries. Explored this with default map-reduce prompts and with refined prompts. Since this method has a downside of splitting the text into chunks with no regard for the logical and structural flow of the text, improved it by adding chunk overlap. This has a drawback of loosing important information from the first chunk. Also, some key points could be longer than the others and we cannot estimate this before.

Another method to improve the algorithm is to pass summary of previous chunk with current chunk through the LLM. This technique progressively improves the summary as it sees more chunks. However, context from the initial chunks could be overrepresented in the overall summary.

To overcome these drawbacks, I finally tried the summarization and topic modeling together in the same algorithm suggested by Isaac Tham [6]. Here, he suggests splitting the summary outputs from one step of the recursive summarization into chunks to be fed into the next step. To achieve this, he suggests clustering chunks semantically into topics and passing these topics into the next iteration of the summarization.

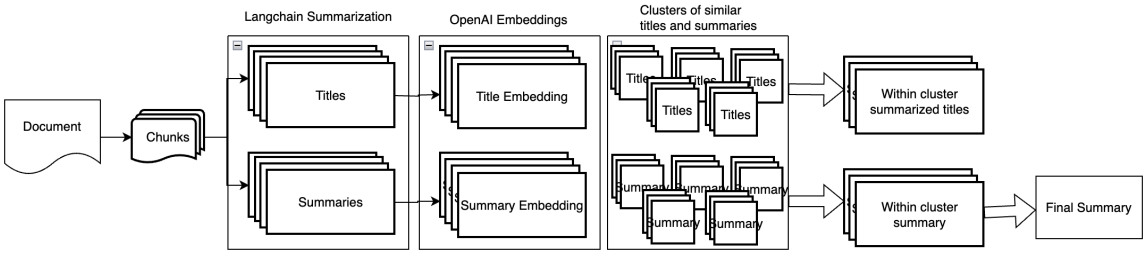

Figure 1: Workflow chart of title and summary extraction algorithm.

3.1 Data Generation and Processing

The text used to summarize is interview transcripts of several pediatric heart surgery patients and their parents after the surgery. I use these interview transcripts to summarize and provide insights about their emotions and medical gaps. The

raw data is split into sentences with each sentence having a minimum of 20 words and a maximum of 80 words as suggested in the article. Instead of creating chunks large enough to fit a context window, the author proposes limiting the chunk size to be the number of sentences it generally takes to express an idea. The author uses 5 sentences with a 1 sentence overlap between chunks to ensure continuity so that each chunk has some contextual information about the previous chunk.

3.2 Extraction of Summary and Title for Each Chunk

As the author suggested, to get the summary as well as title for each chunk I created a custom prompt adapting Langchain's *default summarize chain prompt*. I then create an LLM using *gpt-3.5-turbo-instruct* model to pass the custom prompt with chunk that returns titles and summaries for each chunk.

3.3 Clustering Individual Summaries and Titles into Topics using Embeddings

I use the summaries and titles obtained from the previous step, embed them into 1536-dimension vectors using OpenAI's embeddings. These embeddings capture the semantic meaning of the sentences. I then create chunk similarity matrix for the chunks using these embeddings.

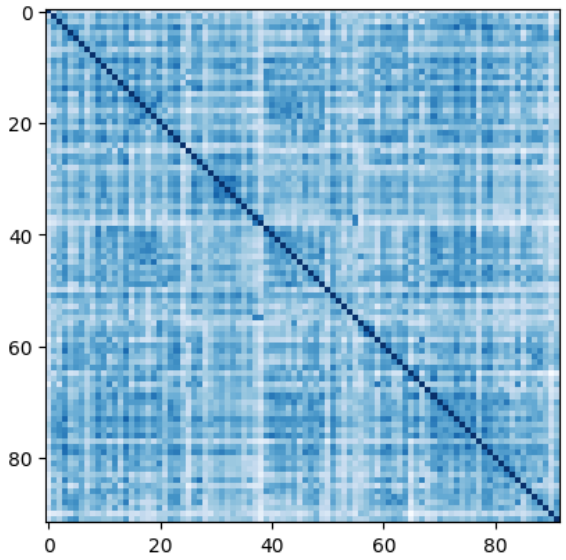

Figure 2: Similarity matrix of all the chunks for patient interview transcript. The darker the color the more similar the chunks are and we can tell groups of chunks that are similar to each other.

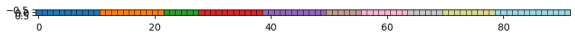

Figure 3: Detected topic clusters for patient and their parent interview post pediatric heart surgery.

Once the similarities scores are generated as shown in Figure 2, use *Louvain Community Detection Algorithm* to detect topics from the chunks. This is because when we cluster chunks into topics, we want them to be semantically very similar to each other and being semantically less similar to chunks in other topics. In Louvain community detection algorithm like the author suggested, I used resolution and proximity bonus hyper parameters to augment similarity score of chunks if their position in the original text is closer to each other. Classification of the chunks can be seen in Figure 3.

3.4 Final Topic Title and Summary

From the last section we have topics and summaries that are semantically coherent. We can now pass these clusters of topics and summaries to the next step of the recursive summarization. And finally to produce at the overall summary of the text, we concatenate the topic summaries and prompt GPT

to summarize them. This workflow can be seen in the workflow chart from Figure 1.

4 Result

There is a clear difference between the results of all the approaches I mentioned here. To summarize, different techniques used performed are map-reduce summarization of text without chunk overlap, map-reduce summarization of text with chunk overlap, map-reduce summarization of text with chunk overlap using custom prompts, passing summary of previous chunk in current chunk and finally title and summary generation of individual chunks in-order to generate final summary. Below are the summarized titles produced for patient interview transcripts using Isaac's approach:

1. Understanding the Emotional Impact of Medical Delivery Gaps in Patients with AAOCA
2. Navigating Emotions and Medical Delivery Gaps in Cardiology
3. Emotional Experiences and Medical Delivery Gaps in Patients with Asymptomatic COVID-19
4. Emotional Turmoil and Medical Delivery Gaps in Critical Care Patients
5. The Emotional Toll of Heart Conditions on Young Patients
6. The Emotional Toll of Surviving a Heart Attack
7. Exploring Patient Emotions and Medical Delivery Gaps in Surgical Treatment for Anomalies
8. The Importance of Early Heart Checks for Children
9. The Emotional Toll of Parenting a Child with a Medical Condition
10. Emotional Support and Medical Delivery Gaps for Patients with Rare Diseases

And the overall summary for the interview transcript used for generating above topics:

This text delves into the emotional toll that medical delivery gaps can have on patients and their families, particularly those with rare conditions. The use of complex medical terminology and acronyms can create communication gaps, leading to frustration and anxiety for patients. These gaps not only affect the quality of medical care but also cause negative emotions such as fear and uncertainty. The importance of clear communication and support from medical professionals is highlighted, as well as the need for more awareness and understanding of rare conditions. The personal stories of patients and their families illustrate the impact of these gaps and the emotional journey they face in seeking proper medical care.

The text also sheds light on the emotional struggles and lack of resources faced by patients with celiac disease, heart conditions, and other rare diseases. It emphasizes the need for improved medical care and support for these patients, as well as better communication and efficiency in the healthcare system. The emotional impact of medical delivery gaps during the COVID-19 pandemic is also discussed, highlighting the importance of being an advocate for oneself and seeking second opinions.

The emotional toll of having a child with a serious diagnosis is also explored, emphasizing the need for a strong support system and better integration of mental health services in medical care. The text also discusses the emotional struggles of patients and their families as they navigate through medical exams and treatments, highlighting the gaps in medical delivery where patients may not have access to necessary support and resources to cope with their emotions.

Overall, this text highlights the significant emotional impact of medical delivery gaps on patients and their families. It emphasizes the need for improved communication, support, and understanding from medical professionals, as well as more awareness and resources for rare conditions. The personal stories shared by patients and their families serve as a reminder of the importance of addressing these gaps and providing proper medical care for all individuals.

The final result summarizes the emotional stress of parents as well as the gaps in the medical delivery system very well. It addresses the questions we wanted to address in the first place. It also segregates the summary into different sections. This method of generating topics and corresponding summaries is helpful not only to retrieve the overall summary of a text but also summary of dif-

ferent subsections. These subsections are defined by the topic headers and we can easily retrieve summary for a particular topic using the topic summaries. For example, if we want to get a summary of *The Emotional Toll of Heart Conditions on Young Patients* we can easily retrieve it and below is the summary.

This text discusses the emotional struggles and medical delivery gaps faced by patients and caregivers of children with heart conditions. It highlights the need for improved medical care and support for young patients, as well as better communication and understanding between medical professionals and families. The lack of information and inconsistent follow-up from doctors causes frustration and guilt for parents, while patients experience fear and worry due to the sudden nature of cardiac events. These emotional experiences shed light on the need for more accessible and comprehensive resources for patients and their families.

5 Conclusion

Automatic text summarization is the process of reducing a long text document into shot text that retains the most important points of the original document. This study explored automated document text summarization using naive LLM techniques as well as advanced text summarization along with titles using LLMs and community detection algorithm proposed by Isaac Tham [6]. The model proposed by Isaac Tham [6] could not only produce an overall summary of a document but also summarize the document text into multiple topics. This helps one extract or use summaries of smaller topics within the text. Text summarization is tricky problem to work on because we cannot analyze performance since we can't compute accuracies of the summaries generated by the algorithm. This can be potentially solved in future by collecting human summaries for a subset of data and using similarity scores between them to compute accuracy and further tweak the prompts. I also think data cleaning and creation of chunks is the most important part of generating appropriate summary. A couple of different things that can improve the results further are, creating a separate chunk for each interview question (this can retain the information and context of a topic which is an interview question in one chunk) to use them as inputs to LLM. And, using interview question and each participant's reply as one chunk so this can retain every information of that idea in one chunk.

6 Reference Links

1. Baeza-Yates and B. Ribeiro-Neto. Modern Information Retrieval. ACM Press, New York, 1999.

2. Y. Gong, X. Liu: Generic Text Summarization Using Relevance Measure and Latent Semantic Analysis. Proceedings of the 24th annual international ACM SIGIR conference on Research and development in information retrieval, New Orleans, Louisiana, United States 2001, pp. 19-25

3. David M Blei, Andrew Y Ng, and Michael I Jordan. 2003. Latent dirichlet allocation. the Journal of machine Learning research 3 (2003), 993–1022.

4. Asli Celikyilmaz and Dilek Hakkani-Tur. 2010. A hybrid hierarchical model for multi-document summarization. In Proceedings of the 48th Annual Meeting of the Association for Computational Linguistics. Association for Computational Linguistics, 815–824.

5. Sumit Asthana, Sagih Hilleli, Pengcheng He and Aaron Halfaker. 2023. Summaries, Highlights, and Action items: Design, implementation and evaluation of an LLM-powered meeting recap system.

6. Isaac Tham. 2023. Summarize Podcast Transcripts and Long Texts Better with NLP and AI